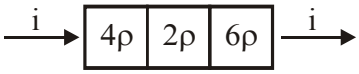
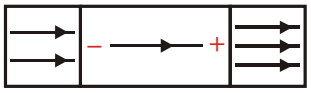
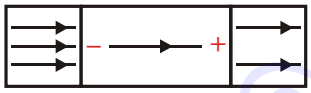
SECTION-I

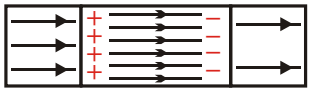
Single Correct Answer Type 4 Q. [3 M (–1)]

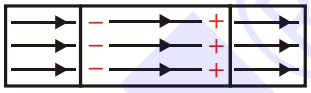
1. Which of the following electric field pattern is correct inside the conductor having three materials of resistivity ?



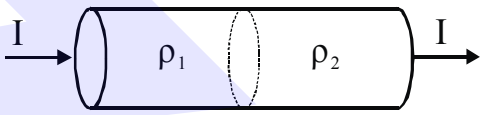
(A) 

(B) 

(C) 

(D) 

2. Two long straight cylindrical conductors with resistivities ρ_1 and ρ_2 respectively are joined together as shown in figure. If current I flows through the conductors, the magnitude of the total free charge at the interface of the two conductors is :-



(A) zero (B) $\dfrac{(\rho_1-\rho_2)I\varepsilon_0}{2}$ (C) $\varepsilon_0 I|\rho_1-\rho_2|$ (D) $\varepsilon_0 I|\rho_1+\rho_2|$

3. A circuit contains a battery of voltage V wired to three resistors of resistance R. Which of the following **cannot** be the power dissipated in the circuit (assuming negligible resistance for the wires)?

(A) $P=\dfrac{V^2}{(3R)}$ (B) $P=\dfrac{3V^2}{R}$ (C) $P=\dfrac{3V^2}{(2R)}$ (D) $P=\dfrac{2V^2}{(R)}$

4. Consider a wire in the shape of a circle carrying a current. Note that as the current progresses along the wire, the direction of current density vector changes in a particular manner, while the current remain unaffected. The agent that is essentially responsible for this is

(A) Source of emf.

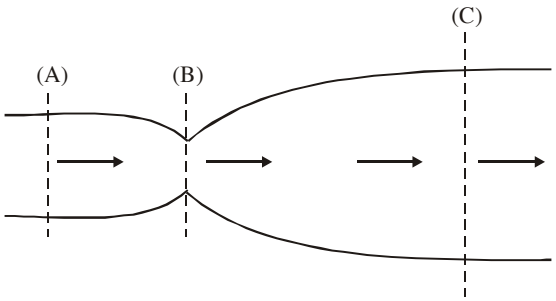
(B) electric field produced by charges accumulated on the surface of wire

(C) the charges just behind a given segment of wire which push them just the right way by repulsion

(D) the charges ahead

Multiple Correct Answer Type 8 Q. [4 M (–1)]

5. Figure shows a copper wire of nonuniform cross section carrying current towards right for sectors A, B and C.



(A) $i_B > i_A > i_C$ (i = current) (B) $E_B > E_A > E_C$ (E = electric field)

(C) $V_B > V_A > V_C$ (V = drift velocity) (D) $J_B > J_A > J_C$ (J = current density)

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6. A metal sphere of radius a is surrounded by a concentric metal sphere of inner radius of b, where b > a. The space between the spheres is filled with a material whose electrical conductivity σ varies with the electric field strength E as σ = kE where k is a constant. A potential difference V is maintained between spheres.

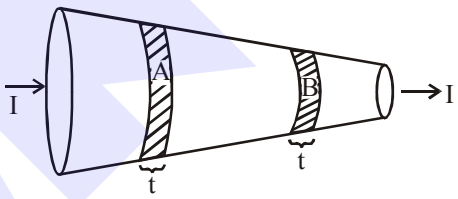
(A) Current is $4\pi r^2 kE^2$, where (a < r < b)

(B) Current is $2\pi r^2 kE^2$, where (a < r < b)

(C) Potential difference between spheres is $V = \sqrt{\frac{I}{4\pi k}}\ell n\left(\frac{b}{a}\right)$ where I is total current

(D) Potential difference between spheres is $V = \sqrt{\frac{I}{2\pi k}}\ell n\left(\frac{b}{a}\right)$ where I is total current

7. Figure shows a homogeneous conductor with decreasing cross-sectional area along the flow of current. Consider two elementary disks A and B of same thickness on conductor as shown. Choose the correct option(s) :-



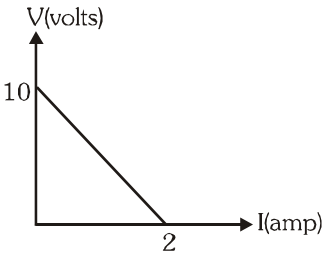
(A) Drift velocity of electrons at B is greater than at A.

(B) No. of electrons per unit volume is same at A and B.

(C) Resistance of segment B is bigger than that of A.

(D) Electric field, current density and resistivity, all are bigger at B than at A.

8. A battery of e.m.f. E and internal resistance r is connected to R. Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current passing through the resistance (I) and potential difference (V) across it. Select the correct alternatives



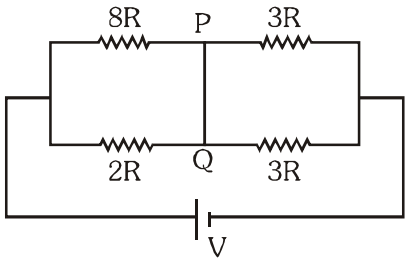
(A) Internal resistance of the battery may be 5 Ω.

(B) Emf of the battery is 10 V

(C) Maximum current which can be taken from the battery is 2 A

(D) V-I graph can never be a straight lines as shown in the figure.

9. Choose correct option(s) corresponding to the given circuit



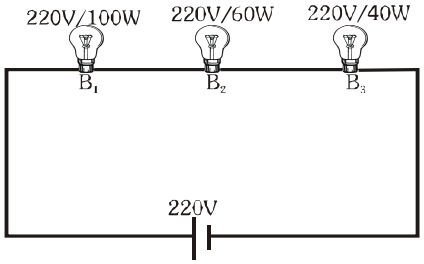
(A) current in PQ will be zero

(B) current in PQ will be P → Q

(C) current in PQ will be Q → P

(D) Equivalent resistance is $\frac{31R}{10}$

10. Figure shows three bulbs B_1, B_2 & B_3. If R, P & I represents resistance, power, and current then choose the ***INCORRECT*** statement.



(A) $R_{B1} > R_{B2} > R_{B3}$

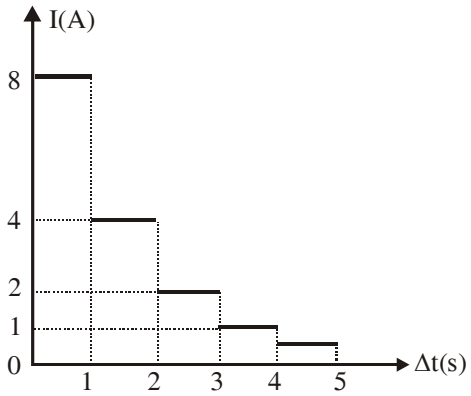
(B) $P_{B1} > P_{B2} > P_{B3}$

(C) $P_{B1} < P_{B2} < P_{B3}$

(D) $I_{B1} > I_{B2} > I_{B3}$

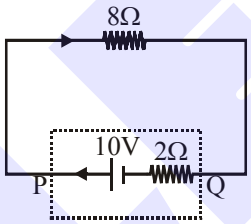
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11. The current in a wire of resistance 10Ω varies as shown.Select correct alternative(s).



(A) The heat produced in the wire in 4 s is 852.5 J
(B) The heat produced in the wire in 4 s is 850 J
(C) The charge flows across any section of the wire in 4 s is 15 C
(D) The charge flows across any section of the wire in 4 s is 15.5 C

12. A battery of emf 10 volt and internal resistance 2Ω is connected to an external resistance 8Ω as shown in the figure :-

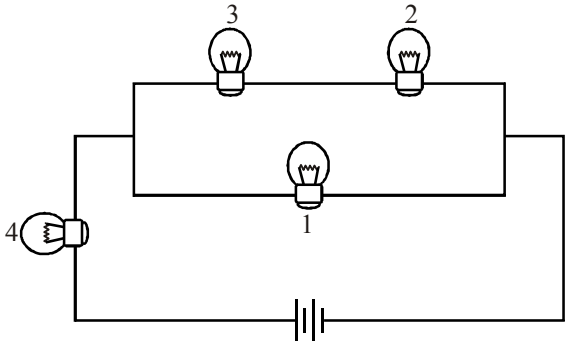


(A) Work done due to conservative electric field while a unit positive charge passes through battery from Q to P (along the arrow) is 8 Joule.
(B) Work done due to conservative electric field while a unit positive charge passes through battery from Q to P (along the arrow) is – 8 Joule.
(C) Work done due to conservative electric field while a unit positive charge passes through 8Ω along the arrow is –8 Joule.
(D) Work done due to non conservative electric field while a unit positive charge moves from Q to P (along the arrow) is 10 Joule.

Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for question nos. 13 to 15

All bulbs consume same power. The resistance of bulb 1 is 36Ω.



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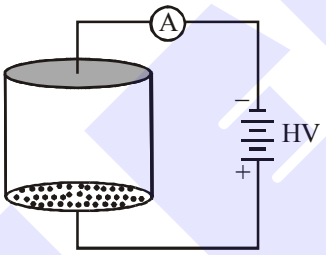
13. What is the resistance of bulb 3 :
(A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω

14. What is the resistance of bulb 4 :
(A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω

15. What is the voltage output of the battery if the power of each bulb is 4W :
(A) 12V (B) 16V (C) 24V (D) None of these

Paragraph for Questions No. 16 and 17

Consider an evacuated cylindrical chamber of height h having rigid conducting plates at the ends and an insulating curved surface as shown in the figure. A number of spherical balls made of a light weight and soft material and coated with a conducting material are placed on the bottom plate. The balls have a radius r <<h. Now a high voltage source (HV) is connected across the conducting plates such that the bottom plate is at $+V_0$ and the top plate at $-V_0$. Due to their conducting surface, the balls will get charged, will become equipotential with the plate and are repelled by it. The balls will eventually collide with the top plate, where the coefficient of restitution can be taken to be zero due to the soft nature of the material of the balls. The electric field in the chamber can be considered to be that of a parallel plate capacitor. Assume that there are no collision between the balls and the interaction between them is negligible. (Ignore gravity)



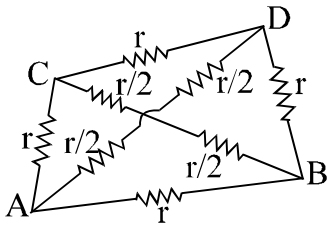
16. Which of the following statements is correct ?
(A) The balls will bounce back to the bottom plate carrying the opposite charge they went up with
(B) the balls will execute simple harmonic motion between the two plates
(C) The balls will bounce back to the bottom plate carrying the same charge they went up with
(D) The balls will stick to the top plate and remain there

17. The average current in the steady state registered by the ammeter in the circuit will be :
(A) Proportional to $V_0^{1/2}$ (B) Proportional to V_0^2
(C) Proportional to the potential V_0 (D) Zero

Linked Comprehension Type (1 Para × 3 Q.) [4 M (–1)]
(Multiple Correct Answer Type)

Paragraph for Question No. 18 to 20

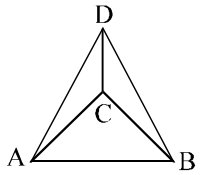
The figure shows a tetrahedron, each side of which has a resistance r

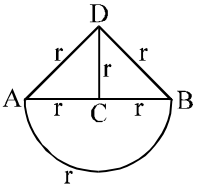


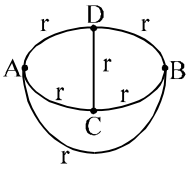
18. Choose the correct statement(s) related to the resistance between any two points.
(A) $R_{AB} = R_{BD} = R_{BC} = R_{CD} = R_{CA} = R_{AD}$ (B) $R_{AB} = R_{AC} = R_{AD} = R_{BD} = R_{BC} \neq R_{CD}$
(C) R_{CD} is the least (D) $R_{AB} = R_{AC} = R_{BC}$ and $R_{CD} = R_{AD} = R_{BD}$

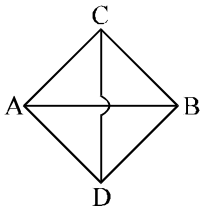
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19. Choose the correct diagram(s), which show two-dimensional equivalent of the tetrahedron.

(A) 

(B) 

(C) 

(D) 

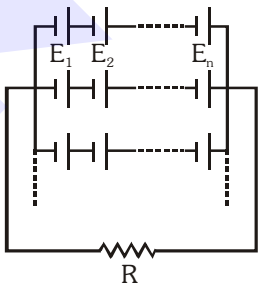
20. If a battery is connected between any two points of the tetrahedron, then identify the correct statement(s).
(A) The potentials of the other two points are always equal.
(B) There always exists a branch through which no current flows.
(C) The current coming out of the battery in each case is same.
(D) None of these

SECTION-II

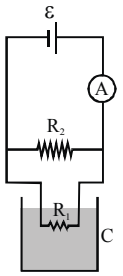
Numerical Answer Type Question **4Q.[3M(0)]**

(upto second decimal place)

21. 324 identical galvanic cells, each of internal resistance 9Ω are arranged as several in-series groups of cells connected in parallel. The arrangement has been laid out so that power output in an externally connected resistance of value 4 Ω is maximum. If n cells are connected in every series group that form parallel combination, then find value of n.



22. The coil of a calorimeter C has a resistance of $R_1 = 60\Omega$. The coil R_1 is connected to the circuit as shown in figure. What is the rise in temperature (°C) of 240 grams of water poured into the calorimeter when it is heated for 7 minutes during which a current flows through the coil and the ammeter shows 3A? The resistance $R_2=30\Omega$. [Disregard the resistances of the battery and the ammeter, and the heat losses and heat capacity of the calorimeter and the resistor and specific heat of water = 4200 J/kg°C]



23. On quadrupling the resistance 'R' connected to a battery the power dissipated in the resistance surprisingly does not change. What is the internal resistance of the battery in Ohm? (Take R = 10Ω)

24. One and a half litres of water at a temperature of 20° C is heated for 15 minutes on an electric table stove burner having two sections with the same resistance. When the sections are connected in parallel, the water begins to boil and 100 grams of it is converted into steam. The latent heat of vaporization is 540 cal per gram. Starting from same initial conditions, the sections are connected in series. In what time (in min) will 100 gm of water get converted into steam. Neglect all heat loses and heat capacity of the container. Density of water 1 gm/cc and specific heat capacity of water = 4200 J/kg°C.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **1 Q. [4 M (0)]**

25. The coil of a heater has resistance of 5Ω and is powered from a source of internal resistance 20Ω. With what resistance (in Ω) should the coil be shunted to reduce the amount of power dissipated in it to one ninth of the value without shunt resistance ?

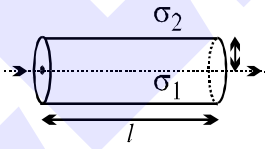
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Subjective Type 5 Q. [4 M (0)]

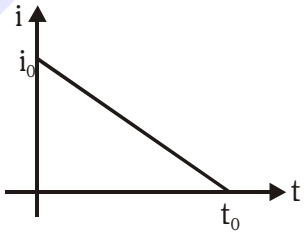
26. A constant current I flows through a manganin wire of resistance R, length *l*, radius of cross-section r, density d and specific heat C. The resistance of manganin wire is independent of the temperature. The amount of heat given off by unit surface area of the wire in unit time is given according to Newton's law of cooling as $\frac{dQ}{dt} = k(T - T_0)$ where T_0 is the temperature of the surrounding. Find the expression of temperature of wire at any instant t.

27. A long conductor of a circular cross-section of radius R is made of a material whose resistivity depends only on the distance 'r' from the conductor's central axis as $\rho = a/r^2$, where a is a constant. The conductor has length L (>>R). Its ends are maintained at a potential difference = ΔV. Find
(a) the electric field and current density in the conductor at any distance 'r' from the central axis.
(b) total rate of joule-heat production.
The rate of loss of heat from the surface of the conductor is $\lambda(T_S - T_A)$, where 'T_S' is the surface temperature of the conductor, 'T_A' the ambient temperature and 'λ' is some constant. Find
(c) surface temperature (T_S) in steady state.
If the thermal conductivity of the conductor's material is 'k' everywhere find
(d) the temperature in the conductor as a function of 'r'. Assuming steady state.

28. A long conductor of circular cross-section has radius r and lenght *l* as shown in the figure. The conductivity of the material near the axis is σ_1. and increases linearly with the distance from axis and becomes σ_2 near the surface. Find the resistance of the conductor if the current enters from the one end and leaves from the other end.

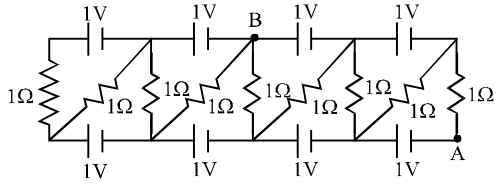


29. Relation between current in conductor and time is shown in figure then determine.



(i) Total charge flow through the conductor
(ii) Write expression of current in terms of time
(iii) If resistance of conductor is R then total heat dissipated across resistance R is

30. Find the potential difference $V_A - V_B$ for the circuit shown in the figure.



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GR # CURRENT ELECTRICITY — **ANSWER KEY**

SECTION-I

Single Correct Answer Type — **4 Q. [3 M (–1)]**

1. Ans. (A) **2. Ans. (C)** **3. Ans. (D)** **4. Ans. (B)**

Multiple Correct Answer Type — **8 Q. [4 M (–1)]**

5. Ans. (B,C,D) **6. Ans. (A, C)** **7. Ans. (A,B,C)** **8. Ans. (A,B,C)** **9. Ans. (C,D)**

10. Ans. (A,B,D) **11. Ans. (B,C)** **12. Ans. (B, D)**

Linked Comprehension Type (Single Correct Answer Type) — **(1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]**

13. Ans. (B) **14. Ans. (A)** **15. Ans. (B)** **16. Ans. (A)** **17. Ans. (B)**

Linked Comprehension Type (Multiple Correct Answer Type) — **(1 Para × 3 Q.) [4 M (–1)]**

18. Ans. (A,D) **19. Ans. (A,B,C,D)** **20. Ans. (A,B,C)**

SECTION-II

Numerical Answer Type Question (upto second decimal place) — **4Q.[3M(0)]**

21. Ans. 12.00 **22. Ans. 25.00** **23. Ans. 20.00** **24. Ans. 60.00**

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) — **1 Q. [4 M (0)]**

25. Ans. 2

Subjective Type — **5 Q. [4 M (0)]**

26. Ans. $T - T_0 = \frac{I^2R}{2\pi r \ell k}\left(1 - e^{-\frac{2kt}{rCd}}\right)$

27. Ans. (a) $\frac{\Delta V}{L}, \frac{\Delta V r^2}{La}$; (b) $\frac{(\Delta V)^2(2\pi)}{La}\left(\frac{R^4}{4}\right)$,

(c) $T_S = T_A + \frac{2\pi(\Delta V)^2}{La\lambda}\left(\frac{R^4}{4}\right)$, (d) $T = T_s + \frac{(\Delta V)^2}{16L^2aK}\left(R^4 - r^4\right)$

28. Ans. $\frac{1}{R} = \frac{\pi r^2}{3l}(2\sigma_2 + \sigma_1)$ **29. Ans.** (i) $\frac{1}{2}i_0t_0$ (ii) $i = i_0\left(1 - \frac{t}{t_0}\right)$ (iii) $\frac{Rt_0i_0^2}{3}$ **30. Ans.** $-\frac{22}{9}$ V

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